Arush Priyank Bhat

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About Me

Physics undergraduate student at Fergusson College (Autonomous). My academic interests lie in astrophysics and cosmology, especially topics like galactic evolution, large scale structures, and dark matter.

Education

Fergusson College (Autonomous) Shivajinagar, Pune
BSc Physics - CGPA: 8.48 *June 2024 – Present*

Projects

Testing the Effect of Inflow Metallicity on Molecular Gas and Star Formation in a One-Zone Gas-Regulator Model Fergusson College (Autonomous)
Python *8th June 2026 - 21st August 2026*

- Formulated and numerically integrated a coupled system of ordinary differential equations (ODEs) in Python (SciPy) to simulate galactic mass and metal conservation over a 10 Gyr cosmic evolution.
- Extended classical b̈athtubg̈as-regulator models by incorporating the Krumholz–McKee–Tumlinson (KMT) framework to link interstellar gas metallicity to dust UV shielding and molecular hydrogen (H_2) formation.
- Discovered a +14.4% transient starburst peak in star formation rate (SFR) triggered by solar-enriched accretion, while proving long-term self-regulation and a 23% cold gas reservoir depletion.
- Validated numerical solver accuracy through strict mass conservation checks (residual $\epsilon_M \sim 10^{-8}$) and convergence tests against theoretical steady-state analytical equilibria.
- Viewable on GitHub

Stellar Population Analysis using Astronomical Data Independent Project
Python, Astropy, Gaia & DESI Datasets *Viewable on GitHub*

- Extracted and analyzed stellar distance, metallicity, magnitude, and temperature metrics from DESI and Gaia survey data.
- Generated Color-Magnitude Diagrams (CMDs) and Hertzsprung-Russell Diagrams (HRDs) for Milky Way surveys to familiarize with complex dataset applications and astronomical structures.

3-Meter Radio Telescope Fergusson College Autonomous & NCRA-TIFR
Motion System Development *Pune, India*

- Contributed to the core development of the motion system for a 3-meter Radio Telescope.
- Collaborated with a team of M.Sc. students on this thesis project, developed in partnership with NCRA-TIFR, Pune.

Skills & Interests

Programming & Tools: Python (Astropy, NumPy, SciPy, Matplotlib), C++, Lua, Git

Curriculum: Classical Mechanics, Statistical Mechanics, Optics, Waves, Oscillations, Mathematical Methods in Physics I&II, Electromagnetism, Multi-variable Calculus, Special Functions and Integral Transforms, Solid State Physics, Atomic and Molecular Physics, Analog Electronics, Numerical Analysis

Interests: Relativity, Early Universe Cosmology, Large Scale Structures, Galactic Structures, Game Development, Robotics, Photography

Extracurricular Activities

Astro Club FC (ICARD) July 2024 – Present
Member *Fergusson College (Autonomous), Pune*

- Planning, management, and volunteering in various Physics outreach events like poster exhibitions and national level seminars.

Volunteer Experience

Poster Exhibition, Astro Club, Fergusson College Jan. 2025 & Jan. 2026

- Contributed to science outreach via creating models and presenting topics on Galactic Rotation Curves (2025) and Applications of Strong/Weak Nuclear force (2026).

National Science Day, IUCAA Feb. 2025

- Contributed to science outreach by presenting the Aditya L1 trajectory model at the National Science Day exhibition.

Awards

International Space Olympiad 2023 Aug. 2023

- Certificate of Excellence in recognition of outstanding performance and successful completion in the Super Senior Category.

Rockschool London - Electric Guitar Grade 1 May 2025

- RSL Level 1 Award presented to students who pass the graded examination in music performance with Distinction.